YEAR 7 · SCIENCE 3.4–3.6 · MATHS 5.1 · EXTRA HOMEWORK 11

How High Does It Bounce?

Tides · bouncing balls · joules · a cooling coffee · angles — all in decimals

Name: _____ **Class:** _____

Date given: _____ **Date due:** Friday 6 November 2026

Why this sheet exists

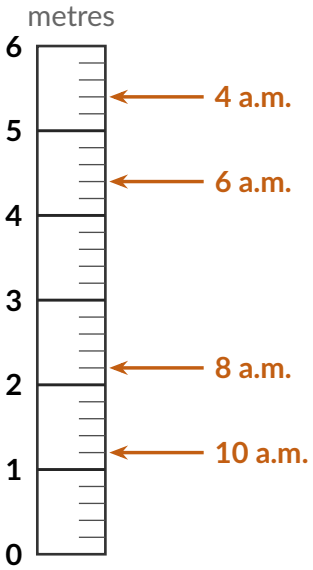
There is **nothing new to calculate** here. Every number is a Unit 4 decimal, and every question is this week's science: the tides, energy, and a ball that will not bounce back to where it started.

The steps are printed as **boxes**. Fill the boxes — that *is* the method. An **orange box** means: write a **0** here.

No calculator. Answer in **full sentences** when it says so.

Part 1 · Tides in decimals

X1 Read the tide board



Word help

tide board	a tall ruler on the harbour wall. It shows the **depth** of the sea.
depth	how deep the water is

Mr Bowen looks at the tide board every 2 hours. Every small line is **0.2 m**. Count in 0.2s: 5, 5.2, 5.4.

(a) Write the depth at each time.

Time	4 a.m.	6 a.m.	8 a.m.	10 a.m.
Depth (m)				

(b) Which time was **high tide**? _____

(c) Which time was **low tide**? _____

X2 The tidal range

Tidal range, from Science 3.4

Tidal range: the difference in depth between high tide and low tide. It is a **subtraction**: high tide − low tide.

Write a 0 in every orange box. Then subtract. **(a)** is done for you.

(a) Mr Bowen's harbour
high 5.4 m, low 1.2 m

	5	.	4	0
−	1	.	2	0
	4	.	2	0

Range: **4.2 m**

(b) Another harbour
high 3.75 m, low 0.9 m

	3	.	7	5
−	0	.	9	0
		.		

Range: _____ m

(c) A third harbour
high 6.1 m, low 2.35 m

	6	.	1	0
−	2	.	3	5
		.		

Range: _____ m

(d) The Bay of Fundy in Canada has the biggest tides in the world. Its tidal range is **16.3 m**. How much bigger is that than the range at Mr Bowen's harbour (4.2 m)?

how much bigger? = the difference = subtract

Line up the points in the grid yourself.

Answer: _____ m

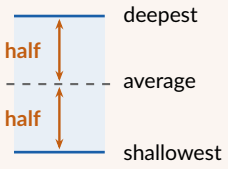
		.		
−		.		
		.		

X3 Half above, half below

The deepest water

The water goes **half the range above** the average depth, and **half the range below** it.

Average 5.0 m, range 2.0 m: half the range is 1.0 m.
Deepest: 5.0 + 1.0 = **6.0** m. Not 7.0 m!



Fill in the table. The first row is done for you.

Average depth	Tidal range	Half the range range ÷ 2	Deepest average + half	Shallowest average − half
5.0 m	2.0 m	2.0 ÷ 2 = 1.0	5.0 + 1.0 = 6.0	5.0 − 1.0 = 4.0
4.5 m	3.0 m	3.0 ÷ 2 =	4.5 +	4.5 −
3.2 m	1.4 m			
6.0 m	2.5 m			

(b) Mr Bowen says: *"In row 2, the deepest water is* 4.5 + 3.0 = 7.5 *m."* What did he do wrong? Answer in a full sentence.

X4 The land has tides too

The land goes up and down too: an **earth tide**. Hanoi goes up and down about **30 cm**, twice a day.

(a) 100 cm = 1 m, so divide by 100. 30 cm = _____ m

(b) Hanoi goes **up** that far twice a day. How far up is that in one day?

0.3 × 2 = _____ m

(c) How far up does Hanoi go in **one week** (7 days)? (It comes down again every time.)

Point out: 6 × 7 = _____ Point back: 0.6 × 7 = _____ m

(d) Nobody in Hanoi feels it. Why not? Answer in a full sentence. Start: **"We do not feel it because …"**

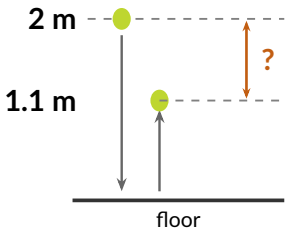
Part 2 · Energy in decimals

X5 Six balls, one floor

Mr Bowen drops six balls, one at a time. He drops each one from **2 m**. He measures how high it **bounces**.

(a) Fill in the last column. The first row is done for you.

Ball	Bounce	How far below 2 m?
tennis ball	1.1 m	2.00 − 1.10 = 0.90 m
ball of clay	0.05 m	2.00 − 0.05 =
bouncy ball	1.75 m	2.00 − 1.75 =
an egg	0 m	2.00 − 0 =
basketball	1.35 m	2.00 − 1.35 =
ping-pong ball	1.5 m	2.00 − 1.5 =



Word help

How far below 2 m? = the difference = **subtract**.
Write 2 as **2.00**. Or count up to 2: 1.6 + 0.4 = 2.

(b) The six bounces were 1.1, 0.05, 1.75, 0, 1.35 and 1.5 m. Write them in order. Start with the **highest**.

_____ _____ _____ _____ _____ _____

(c) At the bounce, some energy changes to **sound** and **heat**.

Which ball changed the **most** energy to sound and heat? _____

Which ball changed the **least**? _____

(d) Finish the sentence.

The lower the bounce, the more energy …

X6 Drop it three times

The mean

Step 1: add up all the values. **Step 2:** divide by how many there are. Three drops: divide by **3**.

(a) The tennis ball bounces **1.12 m**, **1.08 m** and **1.13 m**. Find the mean.

Step 1 add

$$\begin{array}{r} \square 1.12 \\ \square 1.08 \\ +\ \square 1.13 \\ \hline \square\square.\square\square \end{array}$$

Step 2 ÷ 3

$$3 \overline{)\ \square\square.\square\square}$$

Mean: _____ m

(b) The basketball bounces **1.34 m**, **1.38 m** and **1.36 m**. This time, write the digits in the grid yourself. Then write the total in the division.

Step 1 add

$$\begin{array}{r} \square\square.\square\square \\ \square\square.\square\square \\ +\ \square\square.\square\square \\ \hline \square\square.\square\square \end{array}$$

Step 2 ÷ 3

$$3 \overline{)\ \square\square.\square\square}$$

Mean: _____ m

X7 Joules

The book's numbers, from Science 3.5

1 floor of stairs: **2 000 J** run 1 metre: **200 J** boil 1 litre of water: **400 000 J**

(a) The kettle. 1 litre needs 400 000 J, so 0.1 litre needs 400 000 ÷ 10 = 40 000 J. Fill in the table.

Water (litres)	How many 0.1s?	Energy (J)
0.1	1	1 × 40 000 = 40 000
0.2	2	2 × 40 000 =
0.5		
1.5		

(b) Split the number. The first one is done for you.

Climb 2.5 floors: 2 floors + half a floor = 4 000 + 1 000 = **5 000** J

Climb 3.5 floors: 3 floors + half a floor = _____ + _____ = _____ J

Run 12.5 m: 12 m + half a metre = _____ + _____ = _____ J

(c) Mr Bowen's cup of tea needs 0.2 litres of water. How many **floors of stairs** could he climb with the energy that boils it? Use your answer to (a).

_____ ÷ 2 000 = _____ floors

X8 Mr Bowen's coffee goes cold

Mr Bowen makes a hot coffee. Then 30 students ask him questions. The room is **28.5 °C**.

Time (minutes)	0	10	20	30
Coffee (°C)	82.5	64.0	51.5	43.0

(a) How much does it cool in each 10 minutes? Subtract. The first one is done for you.

0 to 10 minutes

	8	2	.	5	
−	6	4	.	0	
	1	8	.	5	

10 to 20 minutes

	6	4	.	0	
−	5	1	.	5	
			.		

20 to 30 minutes

			.		
−			.		
			.		

(b) Each 10 minutes, does it cool by **more** or by **less**? Circle one. more less

(c) Which way does the **thermal energy** move? Tick one.

☐ from the coffee to the room ☐ from the room to the coffee

(d) Mr Bowen forgets his coffee for 3 hours. Will it get **colder** than the room? Circle one: yes no. Then explain in a full sentence. Start: **"Thermal energy only moves from …"**

Part 3 · Some angles

From Maths 5.1

Angles on a **straight line** add up to **180°**. Angles **around a point** add up to **360°**.
Two steps: **1.** Add the angles you know. **2.** Subtract the total from 180° or 360°.

X9 The Earth turns 15° every hour

The Earth spins once a day: 360° in 24 hours. 360 ÷ 24 = 15, so it turns **15° every hour**. Fill in the table.

Hours	Calculation	Degrees	At Mr Bowen's harbour
1	1 × 15	15°	
2	2 × 15		
6			high tide to low tide: a _____ angle
12			high tide to high tide: a _____ line
24			one whole day: a _____ turn
0.5	half of 15		
2.5	30 + _____		

X10 Equal angles around a point

Each one is cut into **equal** angles. Divide 360° by how many there are. Write one digit in each box.

(a) A wind turbine: 3 blades



$$3\overline{)360}$$

= _____°

(b) A ceiling fan: 5 blades



$$5\overline{)360}$$

= _____°

(c) A pizza: 8 slices



$$8\overline{)360}$$

= _____°

(d) A pizza: 16 slices



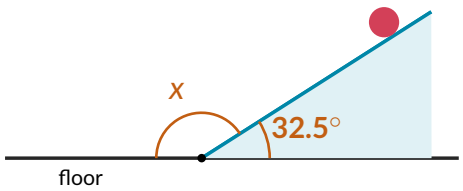
$$2\overline{)45.\square}$$

= _____°

(d) needs no new division: a 16-slice slice is **half** of an 8-slice slice. So it is (c)'s answer ÷ 2.

X11 The ramp

Mr Bowen rolls a ball down a ramp. The ramp makes 32.5° with the floor.



(a) Find x. Write 180 as 180.0.

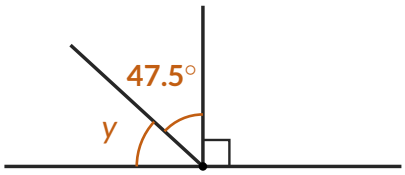
$$\begin{array}{r} 180.\square \\ -\ \ 32.5 \\ \hline \square\square\square.\square \end{array}$$

x = _____°

(b) The ball rolls down. Name the energy change.

_____ ⟶ _____

(c) Three angles on a straight line. Find y in two steps.



Step 1 add

$$\begin{array}{r} \ \ 90.\square \\ +\ \ 47.5 \\ \hline \square\square\square.\square \end{array}$$

Step 2 subtract

$$\begin{array}{r} 180.\square \\ -\ \square\square\square.\square \\ \hline \square\square\square.\square \end{array}$$

y = _____°

X12 The whole sheet in one question

Mr Bowen spends a day at the beach. High tide is **4.6 m** at **7 a.m.** Low tide is **0.75 m** at **1 p.m.**

(a) What is the **tidal range**?

Tidal range: _____ m

(b) From high tide to low tide is _____ hours.

The Earth turns _____ × 15 = _____°.

That is a _____ angle.

(c) Mr Bowen drops a ball from **2 m** onto the sand. It bounces only **0.35 m**. How far below 2 m does it stop?

_____ m

(d) In his hand, what energy store does the ball have?

_____

(e) The ball did not come back to 2 m. Where did the rest of its energy go? Answer in a full sentence. Start: **"When the ball hit the sand, …"**